

Event Density: A Unified Framework for Physics

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Abstract

Event Density (ED) is a substrate ontology whose unifying power comes from one primitive, the **arrow of time** — the irreversible act of commitment — doing load-bearing work across sectors that mainstream physics treats as separate and even as in tension. The same arrow selects the quantum pointer basis (einselection) and *is* gravity’s preferred time (the khronon), so quantum measurement’s “which basis is definite” and gravity’s “which time is flowing” are one primitive wearing two hats, and the problem of time dissolves rather than being reconciled. On that spine, ED is **form-complete across every box a unified framework must check** and **value-inherited exactly where the Standard Model’s free parameters live** — a structural consequence of a pure-relation ontology having no intrinsic scalar, not a gap, and no worse than the Standard Model’s own standing on those numbers.

The report states, box by box and at explicit honesty tiers, what is earned. Quantum mechanics is **re-constructed** from commitment by mapping onto Piron–Solèr (orthogonality reduced to operational

distinguishability, the complex field selected, Born non-contextuality grounded; the residual is the Solèr lattice rigor of an inherited theorem). Gravity is **closed** — the weak-field Einstein metric, the khronometric class with $c_T = c$, $\alpha_2 = 0$ exactly, and the same khronon field giving dark matter (MOND deep field) and dark energy (one Λ), with a new healthy-branch bound $\lambda < 1/3$, $\lambda \neq 1$ placing ED off general relativity. The gauge sector’s **shape is derived** — $SU(N)$ form, non-abelian gauging, the F^2 action, the mass-gap mechanism, and a single hypercharge (the arrow wearing a phase) — the structure the Standard Model postulates wholesale. Charge is **native** (quantized winding, an integral Gauss law), with the smooth electromagnetic field a coarse-grained shadow. Chirality is a **theorem**: the parity-clean substrate is vector for every channel-count, so parity violation is necessarily spontaneous, while *which* force is chiral is inherited representation theory. And the matter sector delivers a **native binding-mass mechanism** — a lone excitation is massless, mass comes from binding, “mass without mass,” the dominant form of real mass — cleanly separated from time dilation.

The report is as loud about its limits as its results. Two structural questions are genuinely open — the gauge multiplicities $\{1, 2, 3\}$ and whether exact charge conservation forces anomaly-freedom on the emergent chiral content — and they reduce to one deep arc (the substrate-to-Dirac reduction) plus one delegatable computation (a linkless-embedding test of the internal dimension), both unsolved in the Standard Model too. One wall is *proven*: the finite-memory substrate cannot reach the primality escape layer, theorem-anchored. The constants are inherited by design. And, decisively, **no distinctive, argument-ending prediction is confirmed yet**: ED makes one confirmed forward prediction and survives the tests that killed rival gravity theories, but the bets that would single it out — a universal decoherence-envelope ratio, a step-like merging-cluster knee, an architecture-independent mass wall, an exactly-zero baryonic Tully–Fisher scatter — are all still open.

The claim, stated at its honest ceiling, is this: one primitive, the arrow of time, recurs as the mechanism across quantum mechanics, gravity, and the Standard Model’s structure, most strikingly where quantum mechanics and gravity were supposed to be irreconcilable. That recurrence is a serious case for a unified framework. It is not yet the empirical case that would end the argument, and this document carries both openly — which is what makes it verification-grade rather than dismissible.

§1 — The Claim and the Bar

The claim

Event Density is a substrate ontology — commitment and participation, with the arrow of time as the single process primitive — advanced here as a **unified framework for physics**. The claim is not that ED reproduces one sector well. It is that one primitive does load-bearing work across sectors that mainstream physics treats as separate: quantum measurement, gravity, gauge structure, charge, chirality, and mass. The report’s job is to state, box by box and at an explicit tier, how much of that is earned and how much is open.

Two words on what this is *not*. It is not a claim to have solved physics — the report carries two genuine structural opens, one proven wall, and no confirmed distinctive prediction, and it says so in every relevant section. And the framing avoids the labels “theory of everything” and “grand unified theory”: the first overclaims, the second technically excludes gravity, and both trigger dismissal for reasons unrelated to content. “A unified framework for physics” is the accurate description and the ceiling of the claim.

The boxes a unified framework must check

A framework earns the word “unified” by accounting for a specific list, and the report is organized around it:

- a minimal foundation (§3);
- quantum mechanics from that foundation (§4);
- gravity, including dark matter and dark energy (§5);
- quantum mechanics and gravity unified rather than glued (§6);
- the gauge forces $SU(3) \times SU(2) \times U(1)$ (§7);
- charge quantization and electromagnetism (§8);
- chirality and parity violation (§9);
- the matter content — spinors, masses (§10);
- the constants (§11);
- internal consistency, i.e. anomalies (§12);
- and, the scientific test, novel falsifiable predictions (§14).

A framework that checks these boxes with derivations, and states honestly where it inherits or stops, is a unified framework in the sense this report defends.

The bar

The evidential standard is deliberately high, and the report’s credibility depends on holding it. Extraordinary claims require extraordinary evidence, and for a unified-framework claim from outside the mainstream, the operative form of that standard is this: **the walls must be as loud as the results**. A report that derived the spine and buried the two structural opens would be dismissible on sight. One that states each result at its true tier, names each open at its true tier, and marks the one thing that is proven impossible, is doing the thing that makes the claim assessable rather than rhetorical.

Concretely, three things are kept rigorously distinct throughout:

1. **What is derived** — a result that follows from the substrate (at the tiers below).
2. **What is inherited** — a value ED’s own logic says it cannot and need not set (the constants, §11), on the same footing as the Standard Model’s free parameters.
3. **What is walled or open** — a limit, either proven (the one primality wall, §13) or a genuine open derivation (#1 and #3), stated as such and not smoothed.

The honesty tiers

Every substantive claim in the report is tagged with one of these tiers, taken from the corpus’s own discipline. They are the vocabulary of the whole document, and no claim is stated above its source’s tier.

- **Derived** — follows from the primitives or the substrate structure (a proof or forced construction).
- **Form-forced / derived-conditional** — the *form* is forced given stated inputs (e.g. the gauge group given the complex amplitude); the conditions are disclosed.
- **Reconstructed** — substantially grounded by mapping onto a standard theorem, with the theorem’s rigor assumed (the quantum-logic keystone, §4).
- **Structural / account** — a grounded argument or identification, not a proof; the unification move (§6) is the report’s central account-tier claim.

- **Measured** — a result from the certified substrate simulator (the binding-mass mechanism, §10; the charge skeleton, §8).
- **Grounded** — a primitive supplies the operational content, conditionally or structurally.
- **Inherited** — a value or structure taken from measurement or standard mathematics by ED’s own logic (the constants; Cl(3,1); Piron–Solèr).
- **Wall** — a proven or honest limit, the order ED does not reach (primality is the one *proven* wall).

The tier word carries the honesty. A result marked “reconstructed” or “account” is not a weaker result dressed up; it is a result stated at exactly the confidence it has earned. The scorecard that follows (§2) tags every box this way, and the sections deliver on each tag.

§2 — The Scorecard

The report’s structure is this table. Read down the “arrow’s job” column and the twelve boxes are not twelve separate achievements — they are one primitive, the arrow of time, showing up twelve times. Read the “tier” column and the honesty is explicit in every row. Every [check] is delivered by its section at the stated tier; every [open] is a genuine open carried openly; every [inherited] is a principled inheritance, not a hole.

Box a unified framework must check	ED status	The arrow’s job here	Tier	§
Minimal foundation	[check] commitment + participation; one process does every job	the arrow <i>is</i> the sole process primitive	Grounded / synthesis	§3
Quantum mechanics from the substrate	[check] reconstructed (orthogonality reduced, $\mathbb{C}$ selected, Born non-contextuality grounded; residual = Solèr lattice rigor)	the arrow selects the pointer basis (einselection) + supplies repeatability	Reconstructed	§4
Gravity (GR + dark matter + dark energy)	[check] closed (GR-I..IV + khronon MOND/DE); +healthy-branch bound $\lambda < 1/3$, $\lambda \neq 1$	the khronon <i>is</i> the arrow made dynamical (preferred time)	Derived (values inherited)	§5

Box a unified framework must check	ED status	The arrow's job here	Tier	§
QM + gravity unified	[check] distinctive — the same arrow is pointer basis <i>and</i> preferred time; the problem of time dissolves	one arrow, two demands	Account (distinctive)	§6
Gauge forces SU(3)×SU(2)×U(1)	[check] shape derived (SU(N) form, non-abelian, F ² , gap, single U(1)); [open] {1,2,3} the wall	the single hypercharge = the arrow wearing a phase	Derived-conditional / Wall (#1)	§7
Charge quantization	[check] native (integer winding, integral Gauss law)	irreversibility quantizes + protects the winding	Measured	§8
Electromagnetism (smooth field)	[check] coherence-weighted limit is Coulomb; the field is a coarse-grained shadow	the arrow sorts the continuum (field = shadow)	Analytic + structural	§8
Chirality / parity violation	[check] operator grounded; theorem: clean substrate vector $\forall N \Rightarrow$ parity violation necessarily spontaneous; [open] casting inherited	the arrow is half of γ^5 ; parity-cleanness forces the vector theorem	Derived (theorem) + inherited casting	§9

Box a unified framework must check	ED status	The arrow's job here	Tier	§
Matter content (spinor, mass)	[check] substrate→Dirac form-complete; native matter/antimatter (C); native binding-mass (“mass without mass”); mass ≠ time dilation; [open] generations/Higgs inherited	the arrow undoubles the spinor + selects matter over antimatter (C) + makes the lone front massless + sets the clock	Form-complete + measured (mass)	§10
The constants (α, mass ratios, Λ)	[inherited] inherited by design — no intrinsic scalar (A1); the SM inherits them too	— (the value layer the arrow does not set)	Inherited-by-design	§11
Internal consistency (anomalies)	[check] conservation + clean-vector baseline solid; [open] chiral cancellation inherited; one candidate T4-gated	clean-vector baseline (arrow) + the worldline-arrow spectral flow	Solid baseline / gated (#3)	§12
Novel falsifiable predictions	[check] ranked weapons + one confirmed forward prediction (UDM law); [open] no argument-ending weapon confirmed	several sharpest weapons trace to the arrow ($\alpha_2=0$, $w=-1$, prohibitions)	Falsifier frontier	§14

How to read the three marks

- **[check]** — ED delivers this, at the tier in the tier column. A parenthetical residual that is inherited standard mathematics (Piron–Solèr) or a rigor-completion does not demote the check; the honesty rides in the tier word (e.g. “reconstructed,” not “proven”).

- **[inherited]** — inherited by design. ED’s own logic (no intrinsic scalar) says it cannot and need not set this. Not a hole, and no worse than the Standard Model’s standing on the same numbers.
- **[open]** — a genuine open. There are exactly two structural ones: the gauge multiplicities #1 (§7) and the chiral anomaly cancellation #3 (§12), which reduce to one deep arc (substrate→Dirac) plus one delegatable tooling task. Plus the honest state of the predictions: none of the argument-ending weapons is confirmed yet (§14).

The scorecard is the skeleton of the report; the arrow column is its thesis; the tier column is its conscience. The sections deliver each row.

§3 — The Arrow of Time: the one process that does every job

The inversion

The microscopic laws are time-reversal invariant; the arrow of time is imposed on them from outside, by a low-entropy boundary condition. That is the standard settlement, and it makes irreversibility a derived, statistical, almost apologetic fact, something the fundamental description does not itself contain.

Event Density inverts it. The arrow is not derived; it is the substrate’s single primitive act. The dynamics is a one-way commitment, P11, and everything reversible in the continuum limit is what survives coarse-graining, not what was there to begin with. This is the opposite of the usual order of explanation, and it is worth being explicit about the cost: ED’s substrate carries *less* symmetry than its own continuum limit, and time-reversal invariance is an emergent artifact rather than a fundamental input.

The payoff of paying that cost is the claim of this report: **sector after sector of physics, worked independently, lands back on that one commitment.** The arrow is not an entry on ED’s list of results. It is the mechanism doing the work in most of them.

What the arrow *is* in ED

In ED the substrate is not a field or a particle or a piece of space. It is a set of relations, chains participating on channels, each carrying an amplitude, and the one thing that happens to those relations is **commitment**: a participation resolves, irreversibly, one way and not the other. That act is the primitive ED calls P11, and it is the arrow of time. Not a clock, not a coordinate, not the ticking of a background parameter. The arrow is the commitment itself: the substrate deciding, once, without taking it back.

Two words carry the whole ontology. **Commitment and participation.** Participation is what is there to be resolved, a chain, on a channel, here, now. Commitment is the resolving. Everything else in this report is built out of that pair, and time is not a third thing added to them. Time *is* the commitment. That is why ED does not need a separate temporal stage: space is a genuine arena the substrate sits in, but time is not an arena, it is the act.

The claim, stated honestly

It is tempting to say “the arrow is the only primitive,” and Allen has arrived at that intuition independently more than once. The honest version is sharper and survives scrutiny. When you audit ED’s primitives one by one against commitment, they sort cleanly into three piles [→ reduction audit]:

- The things the arrow **acts on** (participation, bandwidth, channels, polarity), which it presupposes and cannot be, because you need something there to commit.

- The **arena** it acts in (space's homogeneity, its three dimensions, its scale, the kinds of rules available), which carries no direction of its own.
- The arrow **and every process built from it**: the causal chaining of events, the transport that connects them, the descent that drives them, all forward, all the arrow.

So the precise claim is not that commitment is the only thing. It is that **commitment is the only process**. It is the one source of anything happening at all. The nouns and the stage are presupposed; the verb is singular. That is a weaker statement than “one primitive,” and it is the one that turns out to be true, and it is still remarkable, because a single source of process is exactly what you would need for a unified theory. If everything that *happens* traces to one act, then the sectors of physics are not independent. They are that one act, wearing different clothes.

The jobs — one arrow, sector after sector

What follows is the backbone of the report. Each line is a job the arrow does, and each job is a full section later. Read it as a promise the rest of the document keeps.

Time and irreversibility [→ this section]. The face-value job. The arrow is the direction, the “now” that commits and does not uncommit. Everything else is downstream of this.

The quantum pointer basis [→ §4]. Ask any interpretation of quantum mechanics its hardest question, why measurements come out definite in *this* basis and not some other, and you get silence or hand-waving. In ED the answer is the arrow. A commitment can only resolve in the basis it commits along, the channel basis, so the definite outcomes are definite *because* the arrow selected that basis when it committed. Einselection, which every other account has to install by hand or argue for from decoherence, is simply what the arrow is. ED is a definite-outcome theory because it is a committing theory.

The covering law, the geometry of quantum logic [→ §4]. Once a channel is committed it is locked; ask it again and you get the same answer. That locking is irreversibility, the arrow again, and it is exactly the repeatability that the quantum-logic reconstruction needs to stand its Hilbert space up. The arrow does not just pick the basis; it supplies the reason measurement is repeatable at all.

Gravity's preferred time [→ §5]. ED's gravity is khronometric: it has a preferred foliation of space-time, a “which time is really flowing” that general relativity insists cannot exist. In ED it not only exists, it is forced, because the arrow *is* a preferred direction of time, and a preferred direction is a preferred foliation. The khronon field that carries ED's gravity is the arrow, spread across the substrate.

The unification move [→ §6]. Hold those two jobs next to each other. The pointer basis of quantum mechanics and the preferred time of gravity are not two facts. They are the same arrow. That single identity is ED's answer to quantum gravity, and it is the heart of this report: the two theories were never separate to begin with, they share a spine, and the spine is time committing.

Undoubling the spinor [→ §10]. Put a Dirac field on a lattice and it doubles: you get sixteen species where nature has one. The standard fixes are technical and cost you a symmetry. ED's substrate is a lattice, and it undoubles for free, down to a single species, because the arrow is a genuine direction that breaks the symmetry the doublers rely on. The electron is one electron because the substrate commits.

Matter's very existence [→ §10]. The one chirality-adjacent thing ED gives natively is the matter/antimatter reference, the charge-conjugation label C. It comes from the arrow: first arrival, first

commitment, selects one side. Matter is here rather than annihilated because something committed first.

Parity [→ §9]. The handedness operator of the Standard Model, γ^5 , factors in ED into two pieces, the arrow times a spontaneous choice of orientation. The arrow supplies half of chirality directly. (Which force actually *uses* that handedness is inherited representation theory, and §9 is careful about that line.)

The single hypercharge [→ §7]. The Standard Model has exactly one gauged U(1), one hypercharge, and no one can say why one. In ED there is one because there is one arrow to phase against. Hypercharge is the substrate's polarity measured against the single external flow, and the single external flow is time committing.

Sorting the continuum [→ §3, §8]. Take any smooth law of physics and ask ED which class it falls in, structure-making, structure-erasing, or structure-preserving. The sorter is the arrow. It is what tells Maxwell from diffusion from a conservation law, because the distinction is entirely about how each one treats direction.

Nine jobs, and they are not nine coincidences. They are the pattern you would expect if commitment really is the only process: apparently unrelated structures turning out to be faces of the one one-way act.

Where the honesty lives

Two guardrails, stated here so the rest of the report can lean on them.

First, this is a **convergence, not a theorem**. No single derivation takes commitment and produces all nine jobs from it. What the report has is nine sectors, each worked separately, each landing back on the arrow. That is strong evidence and it is the kind of evidence unification is actually made of, one mechanism that keeps paying out where you had no right to expect it. But it is not a proof that the arrow is logically prior to everything, and this document will not pretend it is. A skeptic is right to press here, and §6 presses back honestly: the force of the claim is the recurrence, not a reduction.

Second, the arrow is **not everything**. It presupposes what it acts on and it needs a stage to act in. ED still carries a spatial arena it does not derive: three dimensions, a scale, a homogeneity, the kinds of rules the substrate can run. Those do not come from the arrow, and later sections that inherit numbers rather than deriving them (the constants, the representation spectrum) are inheriting them partly *because* the arrow is a process and not a stage, and a process cannot hand you the size of the room it runs in. Naming that limit up front is what keeps the spine load-bearing instead of mystical.

What this section buys the rest of the report

From here on, no sector is a standalone result. When §4 reconstructs quantum mechanics, watch for the arrow selecting the basis. When §5 closes gravity, watch for the arrow as the preferred time. When §6 unifies them, it is not introducing a new idea, it is pointing out that the basis-selector and the preferred-time are the same object you met here. The Standard Model sectors, §7 through §10, each open by naming which job of the arrow they are.

The boxes a unified theory must check are the skeleton of this report. The arrow of time is the blood that runs through it. Everything after this page is the same one-way act, seen from one more angle.

§4 — Quantum Mechanics from the Substrate: the arrow selects the pointer basis

The arrow’s job here. Of the roles set out in §3, this section cashes two: the arrow as the quantum **pointer basis** (einselection) and as the source of measurement **repeatability** (the covering law). The claim is not that ED is *compatible* with quantum mechanics. It is that quantum mechanics’ most under-explained structural facts, why outcomes are definite in one basis and not another, and why measurement is repeatable at all, are what a one-way commitment *is*.

The postulate that would not ground

ED’s quantum kinematics (Paper_004) rests on an inner product over the substrate states. In a substrate ontology that cannot simply be assumed; it must trace to the primitives or be flagged as postulated. Two assumptions had blocked three prior attempts: that distinct channels are orthogonal, and that the probability assignment is the Gleason/Born one. The obstruction was always the same shape. Orthogonality was sought as a *metric* fact about vectors, and no ED primitive delivers a metric.

The reconstruction breaks the stall by changing the question, from “are the channel vectors orthogonal?” to “what does the substrate operationally do, and what representation must carry that?” The target is the standard quantum-logic chain: **Piron** (an irreducible, complete, atomistic, orthomodular lattice with the covering law, $\text{rank} \geq 4$, is the lattice of closed subspaces of a Hermitian space over a division ring with involution), then **Solèr** (an infinite orthonormal sequence of equal norm forces the field into $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ and the space to be a genuine Hilbert space), then a physics input to select the field. ED’s lattice is the set of propositions “the commitment resolves in channel-set S ” — the P11 outcomes. The reconstruction maps each axiom to a primitive, with an explicit tier.

Three results

1. Orthogonality reduces to distinguishability. Take candidate channel-states with a *tunable* overlap $c = \langle K|L \rangle$ (do not set $c = 0$). The best confusion-free detection of K , maximizing $\langle K|E|K \rangle$ over POVM elements subject to $\langle L|E|L \rangle = 0$, is the projector onto the complement of $|L\rangle$:

$$\max \langle K|E|K \rangle = 1 - c^2,$$

which equals 1 iff $c = 0$. Perfect distinguishability $\Leftrightarrow$ orthogonality. And ED’s channels *are* perfectly distinguishable from commitment frequencies alone (a pure channel- L state has $b_K = 0$, so it commits to K with frequency zero, P07). So $\langle K|L \rangle = 0$ is not an independent postulate; it is what “perfectly distinguishable by commitment” means. The honest boundary: step (B) is proven *inside* the inner-product representation, so this reduces orthogonality to distinguishability *given* that the representation exists — it does not build the metric from primitives.

2. The complex field is selected. Solèr allows $\{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$; ED picks $\mathbb{C}$ by two standard discriminators, each grounded in a primitive. $\mathbb{R}$ is excluded not because a real space cannot carry a phase (it can, via a Stueckelberg J), but because a real division ring supplies no *central scalar* square root of -1 ; ED’s phase $e^{i\pi}$ is a scalar amplitude factor (Paper_001), not an operator constrained to commute with all observables, so the field must contain i . $\mathbb{H}$ is excluded by the standard tensor-product obstruction: composite systems (the V5 cross-chain joint amplitudes, Paper_063) need a commutative field for $\otimes$ to be well-defined. This is a *selection* from Solèr’s menu, account-tier, not a from-scratch derivation, but it means $\mathbb{C}$ is no longer a bare assumption.

3. The covering law: irreversibility enforces it. The covering law’s operational content is the ideal first-kind measurement, measure an atom, get “yes,” project, and get “yes” again on repetition. The natural worry is that P11’s irreversibility *breaks* the repeatability this needs. It is exactly backwards. Commitment projects the state onto the resolved channel-atom (orthogonally, by P11 exclusivity), and then *locks* it: an irreversible outcome cannot drift, so re-measurement returns the same channel with certainty. **A committed channel is the most repeatable outcome there is.** Irreversibility is not an obstruction to the covering law; it is what supplies it. (Honest scope: this grounds the first-kind core on the *channel* basis; the full lattice-geometric exchange condition and the phase basis remain residual.)

The two arrow-tied resolutions

Two structural questions that other accounts install by hand resolve here directly from the arrow, and they are ED’s distinctive quantum picture.

Einselection is primitive. P11 is the only collapse primitive, and it is a *which-channel* selection. A superposition $|+\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$ is not an intrinsic channel; no primitive collapses to it; committing a $|+\rangle$ state resolves it to $|1\rangle$ or $|2\rangle$, never to $|+\rangle$. So the channel basis is the unique pointer basis, selected by the arrow, not emergent from environmental decoherence. Phase stays a genuine coherence observable (interference is real; a definite phase and a definite channel are complementary), but it is never a commitment basis. **ED is a preferred-basis quantum theory because it is a committing theory.** Ordinary basis-democracy is recovered at the apparatus level (an apparatus is a system whose channels couple to the target observable), so standard QM emerges; the distinctive claim is that einselection is fundamental rather than derived.

Born probabilities are non-contextual. The bandwidth b_K is intrinsic (P04, a bare state property with no apparatus argument), so $P(K) = b_K / \sum b$ is fixed. Because ED commits only in the channel basis, there is exactly *one* substrate commitment context, and Kochen–Specker contextuality (which needs one projector sitting in multiple incompatible contexts) has nowhere to arise. ED assigns non-contextual *probabilities* (Gleason-permitted, the Born rule), never non-contextual definite *values* (Kochen–Specker-forbidden), because outcomes are stochastically committed, not pre-valued. It is consistent with both theorems at once.

Honest status

This is a **coherent reconstruction, not a closed theorem** — and stating that plainly is what keeps it credible. Precisely: - Orthogonality is *reduced* (to distinguishability, given the representation), not produced from primitives; $\mathbb{C}$ is *selected*, not derived; the covering law is *candidate-grounded* on the channel basis only. - The residual is **Solèr lattice rigor**, and it is a *different kind of thing* from ED’s structural opens (§13). The postulate that actually stalled three prior attempts — channel-orthogonality — is **discharged and circularity-audited** (2026-07-08, 2026-07-10), not open. What remains is a rigor-completion: mostly the standard Piron–Solèr machinery (which ED *invokes*, not owes — it sits on the “inherited by design” list alongside Clifford algebra and the lattice-gauge dictionary), plus one genuinely ED-side soft spot named precisely — whether the primitives establish the *full orthomodular identity metric-free* (they give non-distributivity outright; the full identity is the tightest open screw). This is why the box is checked at the tier the word carries: **reconstructed** — substantially grounded by mapping onto a standard theorem, not proven from scratch, and not a missing physics result of the #1/#3 class. - ED’s quantum *logic* (the orthomodular lattice, the covering law) is substrate-exact; its quantum *geometry* (the inner-product metric) is Born-statistical and therefore emergent. The Hilbert

space is exact-as-logic, emergent-as-metric. - The equal-norm (angle) condition, flagged as a circularity in earlier drafts, was discharged operationally in 2026-07-10: in the amplitude representation $\langle K|K \rangle = b_K$, so equal-norm $\Leftrightarrow$ equal-bandwidth, forced by P03 homogeneity.

What is new and solid, independent of the residual rigor, is the **reframe**: orthogonality is operational rather than metric, the complex field is selected by ED's actual phase and composites, and the arrow selects the pointer basis.

What this buys §6

Note what did the work in every result above: the *irreversibility* of commitment. It locked the channel to give repeatability; it made the channel basis the unique pointer basis; it made outcomes stochastic commitments rather than pre-valued facts. That irreversibility is the arrow of §3. Hold onto the specific identification, **the arrow selects the quantum pointer basis** — because §5 will show the same arrow fixes gravity's preferred time, and §6 will argue those are not two facts.

§5 — Gravity: the khronon is the arrow made dynamical

The arrow's job here. In §4 the arrow selected the quantum pointer basis. Here it does something a relativist is trained to say is forbidden: it supplies gravity with a **preferred time**. General relativity insists no such thing exists, that no foliation of spacetime is physically singled out. ED's gravity has one, and it is not smuggled in, it is the arrow of §3 made dynamical. The field that carries ED's gravity, the **khronon**, is the substrate's direction of commitment, promoted to a degree of freedom. Gravity, in ED, is what the arrow looks like at large scale.

This is the sector where ED is simultaneously most complete and most distinctive. It reproduces general relativity where GR is tested, and it departs from GR exactly where the arrow leaves a fingerprint.

The weak field is Einstein's

The lapse of the emergent metric is set by the substrate bandwidth, $N^2 \sim b$, and carrying that through with the correct factor of two yields the **weak-field Einstein metric** and the three classical tests (perihelion precession, light bending, Shapiro delay) at the standard values (GR-I, Derived). Newton's constant comes out as $G = c^3 \ell_P^2 / \hbar$ once the substrate grain is identified with the Planck length (Paper_027; ℓ_P inherited, the relation derived). So at the level everyone actually measures, ED gravity is the textbook metric. Nothing exotic is on offer in the regime where GR is confirmed, which is the correct behavior for a theory that has to survive the Solar System.

The class is forced, and it is khronometric

What kind of gravity is it, structurally? Lovelock's theorem plus a mode count fixes the answer: ED gravity is in the **khronometric class**, two tensor (graviton) modes plus one scalar khronon (GR-II, Derived/Grounded). The tensor sector propagates at the speed of light, $c_T = c$, *structurally* rather than by fine-tuning, so ED is clean against the GW170817 bound that killed a swathe of modified-gravity models. The Lorentz violation the khronon carries is universal and confined to the safe sector.

The dynamical rule for the khronon field is $\ddot{b} = D\nabla^2 b - \kappa\rho$, built and run as a simulation (GR-III, Derived/Measured): it has the Newtonian fixed point, gives $r_s \propto M$, produces a frozen $b \rightarrow 0$ horizon with an area law, and fixes the khronon speed $c_s = c$. And the preferred-frame parameters, the usual

place a preferred-time theory dies, are safe: $\alpha_2 = 0$ **exactly**, and $\alpha_1 = -4\lambda_{\text{local}}$ with $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$, safe by at least seventy orders of magnitude because becoming is sparse (GR-IV). The exact $\alpha_2 = 0$ is not a fitted escape; it is a structural prediction, and a sharp one.

One field: gravity, dark matter, dark energy

The reason to care about this sector is not that it re-derives GR. It is that the *same* khronon does three jobs that mainstream physics assigns to three unrelated things.

- In its **high-acceleration** face, it is Newton and the Einstein metric (above).
- In its **deep field**, it is the dark-matter phenomenology. KM-I embeds MOND relativistically as the khronon's low-acceleration behavior, forced given the ED combination rule $a = \sqrt{a_N \cdot a_0}$ (Paper_030), with the transition scale $a_0 = cH_0/(2\pi)$ coming from the cosmic decoupling surface (Paper_029), the slope-4 Baryonic Tully–Fisher relation $v^4 = GMa_0$ with zero intrinsic scatter (Paper_031), and the AQUAL field equation (Paper_036). The lensing, ghost, and PPN kill-checks pass (KM-I, Grounded/conditional). ED does not add dark matter; it reads the rotation curves off the same field that gives Newton.
- In its **cosmological** face, it is the dark-fluid and a single cosmological constant Λ (KM-II; the Λ *magnitude* is inherited, targets #10).

That is the payoffline of the whole sector, and it is worth stating without hedging: **gravity, dark matter, and dark energy are one field in ED, and that field is the arrow.** (Paper_OneField_Letter is the capstone.)

The new bound: ED is on the healthy branch, not GR

One quantitative piece was tightened this session. The khronometric scalar's kinetic structure has the form $K_{ij}K^{ij} - \lambda K^2 = \text{shear}^2 + (1/3 - \lambda)K^2$. Ghost-freedom requires the low branch, $\lambda < 1/3$. General relativity is the special point $\lambda = 1$, and GR escapes the conformal-mode ghost using full four-dimensional diffeomorphism invariance. **ED cannot use that escape**, because the arrow makes the khronon (the expansion/conformal mode) genuinely propagate, breaking the 4-diffeomorphism symmetry GR leans on. So health *forces* ED onto the low branch: $\lambda < 1/3$, and in particular $\lambda \neq 1$, ED is not GR (form-forced bound; the exact value of λ in $[0, 1/3)$ is F-dependent and inherited). This is a distinctive, falsifiable structural claim: ED lives on the Hořava *healthy* branch, and its scalar sector is not general relativity's.

Where ED departs from GR (the falsifiable edge)

Collecting the fingerprints, because these are the sector's weapons, not caveats: - **ED gravity is MOND, not Einstein, in the deep field**, and the emergent metric is kinematic rather than a dynamical Einstein field (the nonlinear completion is the interference cross-term, not the Einstein–Hilbert term). This is a genuine departure, and it is the source of the dark-matter phenomenology rather than an add-on. - $\alpha_2 = 0$ **exactly** — a hard preferred-frame prediction. - $\lambda < 1/3, \lambda \neq 1$ — the healthy-branch bound. - **Weak-lensing depends on substrate activity** (Paper_038.6), and the **merging-cluster offset-velocity “knee”** is experiment-ready (Paper_OffsetVelocityLaw). These are on the falsifier frontier (§14).

Honest debts

Stated plainly, because a closed sector still has edges: - **Clusters and the CMB are owed.** KM-II carries a clustering regulator dial that reproduces the phenomenology but is not yet pinned from first principles; the cluster-scale and CMB fits are the standing open debt of the line. - **Λ 's magnitude is inherited**, not derived (the structure, sign, and scaling are grounded; the number is the $\theta_{ED} \approx 10^{-122}$ inherited value). - **The exact λ is inherited** (the bound is derived; the point in $[0, 1/3)$ is not). - **GW polarization content** beyond $c_T = c$ is viable but its exact couplings are not fully derived, the one live tension, not a failure.

None of these demote the sector. The GR quartet is closed and Derived; the one-field unification of dark matter and dark energy is grounded and distinctive; the departures from GR are sharp enough to kill the theory if nature disagrees.

What this buys §6

The load-bearing fact to carry forward is the identification made at the top: **gravity's preferred time is the arrow.** The khronon is not a new field standing next to the arrow; it is the arrow, dynamical. Hold that against §4's result, *the arrow selects the quantum pointer basis*, and §6 has both halves it needs. The pointer basis and the preferred foliation are jobs of one primitive. That is the unification, and gravity is where its second half lives.

§6 — The Unification Move: one arrow meets two demands

The previous two sections each ended on the same primitive. §4: the arrow selects the quantum pointer basis. §5: the khronon, gravity's preferred time, is the arrow made dynamical. This section is the observation that those are not two facts. It is the report's central claim, and precisely because it is the most quotable line in the document, it is the one where the tiering has to be strictest.

The demand from each side

Quantum mechanics and general relativity make opposite demands about time, and the conflict has a name: the **problem of time**.

Quantum mechanics wants a preferred time. Unitary evolution runs in a time parameter; the measurement update happens *at* a time and distinguishes before from after; the definiteness of outcomes picks out a basis, and the theory is at its most natural when there is a fact about which time is flowing. General relativity forbids exactly this. Its diffeomorphism invariance says no foliation of spacetime is physically preferred; "which time is really flowing" is, in GR, a meaningless question. Canonical quantum gravity inherits the collision directly, the Wheeler–DeWitt equation famously has no time parameter at all. One theory needs the thing the other prohibits.

ED's stance: side with quantum mechanics, and identify the time

ED does not split the difference. It sides with quantum mechanics: there *is* a fundamental preferred time. It is the arrow, the substrate's direction of commitment (§3). And then it makes the move that turns a stance into a unification: **the preferred time quantum mechanics wanted is the same object gravity's khronon foliation picks out.**

The two demands are met by one primitive. Quantum mechanics wanted a preferred time to make outcomes definite; §4 showed the arrow supplies it, selecting the channel basis. Gravity, in ED, is built on a preferred foliation; §5 showed that foliation *is* the arrow, dynamical. So the preferred frame of the gravity sector and the preferred basis of the quantum sector are not two independent structures that have to be reconciled. They are one structure, wearing the clothes of two sectors. The problem of time does not get solved by reconciling GR's no-preferred-time with QM's need for one; it dissolves, because ED never had GR's version, it has a real arrow, and that arrow is doing both jobs.

That is ED's answer to quantum gravity, stated at the level this report can support: not a quantized metric, but a single primitive that is simultaneously the quantum pointer-basis selector and the gravitational preferred time.

Why this is more than relabeling

The obvious objection is that this is wordplay, calling two things "the arrow" and declaring victory. Three things answer it, and none of them is a proof.

First, ED does not have two arrows to conflate. The reduction audit (§3, and the minimal-ontology paper it draws on) finds commitment to be the *sole process primitive*, the single source of anything dynamical in the theory. Under that finding, the measurement update and the gravitational evolution *cannot* be different arrows, because there is only one process. The identification is not a coincidence noticed after the fact; it is forced by there being one verb.

Second, the single-arrow ontology actually works in both sectors. It is cheap to declare two things identical; it is not cheap for the identification to deliver, on one side, a functioning preferred-basis quantum theory (§4, einselection primitive, Born non-contextuality, the covering law) and, on the other, a gravity that reproduces the weak-field Einstein metric and unifies dark matter and dark energy (§5). One primitive carrying both working structures is the content.

Third, it dissolves a recognized tension rather than papering one over. The problem of time is a real obstacle in mainstream quantum gravity, not a straw man. A framework that makes it *not arise* has done something, even before it has proven anything.

The honesty guard (load-bearing)

Here is the discipline this section must hold, taken directly from the minimal-ontology paper's own warning about this exact claim.

This is a structural synthesis, an account, not a theorem, and its evidential force does not compound. The minimal-ontology paper is explicit: the arrow's many jobs are "a pattern, not a compounding of evidence, seven conditional accounts do not sum to one hard result." The same applies here at the two-sector scale. §4 is a reconstruction (with the Solèr residual); §5 is closed but leans on grounded, not derived, pieces in its dark sectors. Identifying their arrows does not multiply their credibilities into something stronger than either. It exhibits a coherence: one primitive, worked independently in two sectors that mainstream physics treats as unrelated and even in tension, turns out to do a load-bearing job in each, consistently. Coherence-across-sectors is genuine evidence, it is the kind of evidence unification is actually made of, but it is weaker than a derivation, and the report says so.

Two specific limits. (i) There is no single theorem that *forces* "the einselection arrow equals the khronon arrow" from a deeper principle; the identification follows from the sole-process finding, which is itself a P11-*pivoted* audit result (the pivot is a disclosed input, not proven to be the only

possible one). (ii) The identification currently carries no delivered prediction. If the pointer basis and the gravitational foliation are the same object, they should be locked together, and any observable consequence of that locking would be the substrate-scale signature §4 flagged as a candidate, currently unobservable (§14). Until such a signature is found and tested, this is a compelling structural picture, not a confirmed one.

A skeptic is right to press exactly here, and the honest response is not to push back with more assertion. It is to point at the coherence and name its tier: **distinctive, structural, and unconfirmed**.

What this buys the rest of the report

§3 introduced the arrow; §4 and §5 gave its two largest jobs; this section identified them. From here, the Standard Model sectors (§7–§10) are the same primitive again, in smaller and more inherited roles, the single hypercharge it anchors, the continuum it sorts, the spinor it undoubles, the chirality it half-carries. None of those will be a unification on the scale of this one, and none needs to be. The claim of the report is not that the arrow does everything with equal force. It is that one primitive keeps turning up as the mechanism, most strikingly where quantum mechanics and gravity were supposed to be irreconcilable, and that this recurrence, honestly tiered, is the case for calling Event Density a unified framework.

§7 — Gauge Structure: the shape is derived, the multiplicities are the wall

The arrow's job here. The arrow's gauge-sector role is the single hypercharge. The Standard Model has exactly one gauged $U(1)$, and no conventional account says why one. In ED there is one because P09 polarity is one primitive phased against one external flow, the commitment arrow (§3). Hypercharge is the arrow wearing a phase. The rest of this section is the channel structure, and it sets a clean, honest boundary: ED derives the *shape* of the gauge sector that the Standard Model postulates wholesale, and inherits the one piece the Standard Model also does not explain.

What the substrate fixes

Assemble the substrate into a bundle: the base is emergent spacetime, the fiber at each locus is the channel family (P07), the connection is polarity-transport (P05). The gauge group is the fiber's structure group. For a family of N dynamically-indistinguishable channels carrying a complex amplitude $\psi \in \mathbb{C}^N$, the transport that mixes the channels while preserving total bandwidth (a P05 isometry) is the unitary group $U(N) = SU(N) \times U(1)$. So **non-abelian $SU(N)$ is the structure group of the channel fiber** — conditional, and the paper is precise about the condition: indistinguishability *alone* gives only the permutation group S_N ; the continuous $U(N)$ needs the $\mathbb{C}$ -amplitude (from the quantum-logic keystone, §4) and total bandwidth as the sole invariant.

On that structure the rest of the gauge sector's shape follows: - The **gauging is genuinely non-abelian**: position-dependent transport gives non-commuting holonomies, reconciled with einselection because the N channels within a family have no accessible label, so transport is free to mix them while the arrow pins the basis for the $SU(N)$ -invariant observables across families. - The **action is F^2** : the substrate's coherence-deficit on the plaquette holonomy is the Wilson action, $1 - (1/N)\text{Re Tr } U_\square$, which coarse-grains to $-\frac{1}{4}\int \text{Tr}(F^2)$ — the Wilson trace being the matter-averaged per-chain deficit by Schur's lemma, given the effective-action prescription. (The abelian case is grounded on the certified

substrate; the non-abelian lift is analytic.) - The **mass gap has a clean origin**: the same coherence-deficit carries the self-interaction $[A, A]$, which is zero for commuting (abelian) channels and nonzero for non-commuting ones. So in pure Yang-Mills the classical massless flat direction is lifted exactly when the channels are non-abelian — Maxwell is massless because its channels commute, Yang-Mills gaps because they do not. This is the gap's *mechanism*, verified at tree level; the continuum survival (asymptotic freedom, the Clay problem) is a different and harder question, not claimed. - The **single hypercharge** is the arrow's job above: one P09 phase against one arrow gives one $U(1)_Y$.

Put together: $\prod_i SU(N_i) \times U(1)_Y$. The Standard Model takes $SU(3) \times SU(2) \times U(1)$ as a postulate. ED derives that this is the *kind* of object the channel substrate produces — non-abelian, F^2 -acting, gap-carrying, with a single hypercharge — given its amplitude structure. That is a real accounting of structure the Standard Model simply assumes.

The wall: why {1, 2, 3}

What ED does not do is select the multiplicities. The framework gives $SU(N)$ for any N ; it does not say why the stable channel-families are singlets, doublets, and triplets and stop there. This is the report's largest genuine open structural question (#1), and three things should be said about it plainly.

First, ED's natural route to it *failed*, and that is reported as a failure, not smoothed over: channel-family stability does not select {1,2,3} — a symmetric multiplet is stable for all N (the binding energy grows with N , the Hessian stays positive). ED does not currently derive the multiplicities.

Second, the Standard Model does not explain them either. The gauge group and its representation content are inputs there. So ED's status on this specific question — the shape derived, the multiplicities inherited — is not *worse* than the Standard Model's; it is the same inheritance, on top of a derivation of the surrounding structure the Standard Model does not attempt.

Third, the open question is bounded and well-posed, not a vague deficiency. It reduces to a single sharp question — why the internal channel-amplitude dimension is three — and the one live route to it (a 3D-linking argument that would tie the internal dimension to the spatial one) turns on a standard, non-ED-specific mathematical task: detecting whether the substrate's participation graph is intrinsically linked, which requires an established linkless-embedding-detection algorithm rather than any new ED insight. That is a tooling task that someone with the right computational geometry tools could run; it does not require the theory to say anything further. So #1 is an open problem with a delegatable next step, held honestly as unbuilt.

What is inherited or gated

- **Couplings and scales** (g , the effective graph scale a) set the normalization; inherited.
- **Charge values** (per-particle hypercharge) are the per-channel P09 advance rates; inherited. The single $U(1)_Y$ is grounded, but its charge normalization is not derived.
- **Electroweak breaking** ($SU(2) \times U(1)_Y \rightarrow U(1)_{EM}$, the Weinberg angle, the W/Z masses) rides on the substrate-Higgs and is Higgs-gated. The unbroken group is derived; the breaking is not.

Scope

- $SU(N)$ is derived-conditional on the $\mathbb{C}$ -amplitude, not from indistinguishability alone (which gives S_N).

- The F^2 action's non-abelian form is at the gauge-program (analytic) tier; only the abelian case is simulator-grounded.
- The mass gap is a mechanism, not a Yang–Mills existence proof; continuum survival is open.
- The single hypercharge is a grounded identification (P09 is one primitive), not a forced collapse.
- Spin-SU(2) (the spacetime frame bundle) is a separate object, not this internal gauge structure (§10 handles the spinor).

What this buys the report

This section plants the wall the whole Standard Model quarter refers back to. The inherited casting of §9 (which force is chiral = SU(2) pseudoreality), the inherited charge magnitudes of §8, and the inherited generation spectrum of §10 are all the *same* rep-spectrum wall, #1, and it lives here: ED derives the gauge sector's shape and inherits its multiplicities. Framed correctly, that is one bounded, delegatable open problem sitting underneath a genuine derivation of structure — not three separate gaps, and not a vague failure.

§8 — Charge and Electromagnetism: a native skeleton, a shadow field

The arrow's job here. Two of the arrow's roles from §3 show up together in the charge sector. Irreversibility (P11) makes the substrate's phase winding single-valued, which is what quantizes charge and protects it. And the arrow's continuum-sorting job decides what electromagnetism *is* in ED: not a fundamental field, but the coarse-grained shadow the determinate substrate casts.

The skeleton is native and quantized

ED's charge result is reached graph-first: the participation graph is asked which topological invariants it carries, and only then is the survivor compared to charge. The survivor is the U(1) holonomy on cycles. Committed single-valued polarity (P09 phase made single-valued by P11 commitment) forces its holonomy to an **integer winding** $w \in \mathbb{Z}$ — exactly $\pi_1(U(1)) = \mathbb{Z}$, to machine precision. That winding is **conserved and irreversibility-protected**: changing it would require uncommitting a polarity, which P11 forbids. And it sources a field topologically, obeying the **integral Gauss law** — the circulation around any enclosing loop is $2\pi w$, independent of the loop's size and shape (B4).

So charge quantization, conservation, and the unscreenable Gauss-law sourcing are structural facts of the ED graph, not inputs. Quantization is $\pi_1(U(1)) = \mathbb{Z}$; protection is the arrow. This is the topological skeleton of charge, and it is genuinely native.

The scope of this result is the skeleton, not the values. ED produces an integer winding with no selected magnitude — not the $\pm 1, \mp \frac{1}{2}$ spectrum, not the fine-structure constant. Those live in the representation content, which the report inherits (§7, §13). §8's claim is about the *structure* of charge: that it is topological, quantized, and protected. The magnitudes are a separate question, and ED does not set them.

The smooth field is the coherence-weighted limit — and a shadow

B4 established the skeleton but left one question open: the determinate substrate carries the integer winding and the Gauss law, but it does not produce a determined local $1/r^2$ field. The per-edge configuration is gauge/sweep-dependent; a determined isotropic Coulomb field appears only if one removes

both orientation-blindness and P11 — at which point the system is ordinary lattice-field relaxation and no longer ED. B4’s open edge was whether the *coarse-grained* limit recovers Maxwell.

It does. The Maxwell action is already latent in the coherence term: $\cos^2(\Delta\phi/2) \approx 1 - \frac{1}{4}(\nabla\phi)^2$, so the incoherence is the standard $\frac{1}{4}(\nabla\phi)^2$ electrostatic action. Minimizing that coherence action around a point charge gives the **Coulomb field** — the FFT Poisson solve fits $\phi \sim A/r$ with $p = 1$ (3D Coulomb) at $R^2 = 0.97$, best among the tested exponents, with $\phi \cdot r \approx \text{const}$ in the near field (Paper_MaxwellEmergentShadow, this session). So the coherence-weighted continuum limit of B4’s holonomies is Maxwell/Coulomb.

But that is a statement about the coherence-weighted ensemble, not about the determinate substrate. The certified substrate is kinetic and committal — it traps and commits, it does not relax toward a field-minimizing configuration (Paper_Continuum). It is Σ -blind to the phase sector. So the determinate substrate does **not** cast the smooth field: B4 §7 resolves negatively for it. The smooth Maxwell field is the thick-limit **shadow** — form native (the charge skeleton is real substrate structure), smooth field coarse-grained (a came-back-no for determinate-dynamics, the same verdict ED reached for the diffusion PDE).

ED’s monist position on electromagnetism

Put the two halves together and the sector has a clean ontology. The charge *skeleton* — winding, quantization, Gauss law — is native substrate structure. The electromagnetic *field* — the smooth $A_\mu, F_{\{\mu\nu\}}$, the Coulomb $1/r^2$ — is not a second fundamental thing sitting beside the substrate; it is what the substrate looks like coarse-grained, as the smooth metric (§5, emergent and kinematic) and the diffusion equation are — the determined continuum is always the coarse-grained object, never a second primitive. ED does not have a fundamental field and a fundamental substrate to reconcile. It has one substrate, and the field is its shadow. That is the same monist move the whole program makes, and electromagnetism is its cleanest instance, because the lattice-to-continuum dictionary here (Wilson’s lattice gauge theory) is already standard.

Scope

- The winding is a structural realization of charge’s skeleton. It is not a claim that the winding *is* electric charge, and no electromagnetic *content* is derived.
- The charge magnitudes and the fine-structure constant are not produced (inherited; the spectrum question is §7/#1).
- The skeleton results are on minimal certified arenas (rings, small grids); multi-winding interactions and coupling to matter fronts are not covered.
- The shadow result is analytic (the coherence-action minimizer is Coulomb) plus the structural reading (the determinate substrate does not cast it). It is not a certified-substrate simulation of the field, because the phase sector is Σ -blind, so there is nothing to run there.

What this buys the report

This is the arrow in the charge sector: irreversibility quantizing and protecting the winding, and the continuum-sorting deciding that electromagnetism is a shadow rather than a fundamental field. It also sets the pattern for §5’s metric and the rest of ED’s continuum objects — the determined continuum is always the coarse-grained shadow, and charge is where that reading is sharpest. The representation

content that would fix the charge *magnitudes* is the open rep-spectrum question, taken up in §7 and §13.

§9 — Chirality: ED builds the stage, inherits the assignment

The arrow's job here. Chirality is where the arrow does a half-job, and the honesty of the section is in saying which half. The chirality operator γ^5 factors, in ED, into the arrow times a spatial orientation, so the arrow literally *is* one of its two ingredients. And the arrow's parity-cleanness — the substrate has no reflection in its toolkit — is what forces the central theorem: the clean substrate is vector, and any parity violation must be spontaneous. What the arrow does *not* do is pick which force is chiral. That is inherited.

Parity violation, the fact that the weak force couples to left-handed fermions and not right, is one of the starkest facts in the Standard Model, and it is written into the law by hand. The question for ED is whether the substrate produces it or inherits it. The answer is precise: ED builds the entire stage on which chirality lives and inherits the one assignment.

The operator is grounded

The Dirac chirality operator is $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Read its two factors in ED. γ^0 is the arrow: ED's timelike direction is the commitment order (P11, retarded transport). $\gamma^1\gamma^2\gamma^3$ is the oriented volume element, a handedness of the spatial 3-frame that flips sign under reflection. So γ^5 is the arrow times a spatial orientation, both factors native. This grounds an operator the earlier parity work had found missing, and it explains why that work found “no local screw”: γ^5 is not a per-channel object at all, it is global, the one arrow times the one global orientation.

The operator is only as real as its orientation factor, and two facts make it real. The orientation is **spontaneous** (derived): ED's rules are parity-symmetric, no primitive is a reflection, and a parity-symmetric rule set cannot fix a handedness, so any orientation the substrate carries is chosen by symmetry breaking, like a ferromagnet picking a direction. And the emergent space is **orientable** (structural): building a non-orientable space requires a reflection somewhere in the frame bundle, and ED has no reflection, so all frame relations are proper rotations and a single orientation propagates across the causal patch. The same one fact, no reflection primitive, does both jobs: it makes the orientation spontaneous and makes the space able to carry it globally.

The theorem: the clean substrate is vector

Model a family of N channels as a commitment map on an emergent chain, $H(k) = e^{ik}A + e^{-ik}B$, with A the forward (arrow-directed) hop, B the backward hop, k the wavenumber. Net chirality is the **point-gap winding** of this map, the number of times $\det H(k)$ encircles a reference as k runs its period; nonzero is chiral, zero is vector.

Parity-cleanness fixes B . Parity is the spatial reflection $k \rightarrow -k$ together with the lane reflection S of the channel row, and parity-symmetry of H forces the backward hop to mirror the forward one, $B = S A S^{-1}$. Then

$$S H(k) S^{-1} = e^{ik}B + e^{-ik}A = H(-k),$$

so $\det H(k) = \det H(-k)$: the determinant is **even** in k . An even determinant retraces its own path over the period, encircling nothing, so the **winding is identically zero, for every N and every**

forward hop A. This was checked directly for $N = 1$ through 6 over many random hops, zero in every case.

The parity-clean substrate carries no net chirality at any channel-count. It is vector for one channel, two, three, all of them.

The corollary is the strong positive: since the clean transport is vector, any parity violation must break the parity symmetry, which means it must be the spontaneous orientation choice. **If parity is violated in ED, its direction is necessarily spontaneous** — not a handedness written into the law, because the clean rules provably cannot carry one, but a contingent, symmetry-broken choice. This is what turns the earlier verdict from “we looked and did not find chirality” into “the clean rules cannot carry it, and here is the proof.”

Capability is not selection

Having a defined γ^5 is not yet parity violation; the dynamics has to treat the two handednesses differently. That splits into capability (is there a genuine left and right to couple to?) and selection (does the dynamics use it?). The capability structure is clean and is Allen’s highway picture made precise: a parity-clean coupling that references the orientation must live in the reflection-odd part of the gauge structure. For $N = 1$ (abelian) that sector is empty, so the coupling is **forced vector** — which is electromagnetism. For $N \geq 2$ it is nonempty, so a chiral coupling is **possible**. The geometry says whether the highway has a left and a right lane; it does not say whether the universe drives in one. That is the whole content of the wall.

The casting is inherited

Which channel-count actually ends up chiral, once the symmetry is broken, is not a transport fact. The clean winding is $\emptyset$ (the theorem); under a natural breaking it is N , monotone in the channel-count. The Standard Model’s pattern is **non-monotone**: vector, chiral, vector for one, two, three channels. It matches neither $\emptyset$ nor N . So the transport dynamics does not select it.

Where the selection lives is representation theory: the fundamental of $SU(2)$ is **pseudoreal**, while the fundamentals of $SU(N \geq 3)$ are complex. “Two channels is special” is a group-theoretic fact, not a parity-of- N fact, and it is the genuine invariant behind the lane picture’s correct hit on the weak force. ED inherits it, exactly as it inherits the gauge multiplicities and the constants (this is the same representation question as §7 and §13). The contribution here is to name precisely what is behind the wall.

The prediction, and the anomaly baseline

Because the clean substrate is provably vector, ED’s parity violation is necessarily a spontaneous, per-universe orientation tied to the arrow, and the same arrow sets the matter/antimatter sign. So ED predicts the **gauge handedness and the matter/antimatter sign are correlated** — two faces of one first-commitment choice — against the Standard Model’s fixed, uncorrelated, law-level handedness. The claim is theoretically firm (it rests on the vector theorem, not a stance), but its **testability is open**: the corpus register flags it as not testable with current data, so §14 carries it as a distinctive-but-not-yet-testable prediction rather than a live weapon.

The theorem also fixes the anomaly baseline: a vector theory is automatically anomaly-free, so the clean substrate carries no gauge anomaly (at the transport level, contingent on the substrate-to-Dirac

descent). The Standard Model’s nontrivial chiral cancellation is a property of the inherited chiral content, so it is inherited with that content (§12).

Scope

- The casting is inherited, not derived. The wall is strengthened and explained, not breached.
- The theorem is at the channel-transport level; the relativistic descent to the full Dirac sector is the standing open arc (§10, §13).
- No representation spectrum, hypercharges, masses, or charge magnitudes.

What this buys the report

Chirality is the arrow doing half a job cleanly: it *is* half of γ^5 , and its parity-cleanness forces the vector theorem, so parity violation is provably spontaneous rather than law-level. The other half — which force is chiral — is the representation-theoretic inheritance that also drives §7 and §12. And the correlated-handedness prediction is a distinctive claim (testability open, §14). The verdict is a clean division: ED builds the stage of chirality and inherits the one assignment.

§10 — Matter: the spinor, and mass without mass

The arrow’s job here. The matter sector uses the arrow four ways, and this section is organized around them. The arrow **undoubles** the lattice spinor, cutting the Nielsen–Ninomiya sixteen species to one. Its first commitment **selects matter over antimatter** (the charge-conjugation reference C), the one chirality-adjacent thing ED fixes natively. The arrow makes a lone excitation **massless**, moving always at the substrate speed, which is what forces mass to come from binding. And the arrow’s commitment rate sets a **clock**, which turns out to be time dilation and not mass, a distinction the substrate had entangled and this section separates.

The matter sector splits into two questions: does ED produce the Dirac spinor, and does it produce mass? The spinor is form-complete and structure-inherited, the standard ED shape. Mass is the more interesting result, because ED at first looks unable to make it at all, and then makes it exactly where most of it actually lives.

The spinor: substrate to Dirac, form-complete

The Dirac operator’s *form* is forced: it is the unique first-order Clifford-linear factorization of the substrate’s Klein–Gordon operator, $D = i\gamma^\mu \partial_\mu - mc/\hbar$. Its *substrate continuum limit* is now computed, not asserted: retarded transport on the participation graph gives $D(p) = \sum \gamma^\mu (e^{ip_\mu} - 1) \approx i\gamma^\mu p_\mu$ near $p = 0$ (Paper_106, corrected this session from an earlier “asserted via inheritance” status). And the **arrow undoubles it**: a naive Hermitian lattice Dirac operator has sixteen doublers, and the arrow’s retardation is exactly the Wilson term that removes them, leaving a single species at the origin, verified directly (chiral_3p1d.py). This is ED’s concrete escape from Nielsen–Ninomiya, a relational graph plus a genuine (non-Hermitian) arrow.

The single 3+1D survivor is a full, vector-like Dirac point (both chiralities), which is exactly right: the Dirac fermion *is* $L \oplus R$, and its handedness is the separate weak-force question settled in §9. So the “vector-like survivor” is a feature of a correct Dirac reduction, not a gap.

What is inherited is the standard ED value/structure layer: the $Cl(3,1)$ spinor structure (via the acoustic-metric signature, the continuum arc), the metric signature, and the mass value. The one route that would make the spinor fully substrate-native, building the 4-spinor from graph degrees of freedom, runs into the same channel-topology wall as the representation spectrum (§7, §13): canonical channels carry no topology to be a spinor's, and the program that would supply one is unbuilt. So the spinor is form-complete and inherited-structured, with its one deepening blocked at the rep-spectrum wall.

Matter over antimatter

The spinor is vector ($L \oplus R$), so on its own it does not distinguish matter from antimatter. The arrow does. Charge conjugation C is a reference that has to be set somewhere, and in ED it is set by the first commitment: first arrival selects one side, and matter is here rather than annihilated because something committed first. This is the one chirality-adjacent thing ED fixes *natively* (parity, the handedness, is inherited — §9), and it is the same object that appears in §9's operator grounding, where $\gamma^5 = \text{arrow} \times \text{orientation}$ has the arrow as its C -side ingredient and the orientation as its P -side. So C and P are two attributes of one operator, both set at the arrow's early action, and C is the half ED derives. It carries no matter/antimatter *number* (no baryon asymmetry magnitude); it is the reference, native, that the number would be measured against.

Mass looks impossible for a single front

ED's certified rule is **ballistic-or-extinct**: a front advances one hop at the maximal speed or it dies. There is no dwell, no fractional hop, no slow-but-surviving mode. Rest mass needs exactly that missing mode, something that lets a thing sit slower than c . So a lone ED front cannot have rest mass; it moves always at c .

That looks like a failure, and Allen's reframe turns it into the mechanism: a lone front shares the cosmic horizon's defining feature, c -moving, no rest frame, massless. Mass is therefore not a property a front can carry. If mass exists in ED, it has to come from what a *collection* of fronts can do that one cannot, bind. And this is how most real mass already works: roughly ninety-nine percent of a proton's mass is binding energy, not the Higgs coupling; a box of light has rest mass though every photon in it moves at c .

Mass from binding, measured

The test is whether ED's cross-chain coupling V_5 , in its known finite-reach retarded attractive form, confines massless fronts into a bound composite. On the certified simulator it does (Paper_MassWithoutMass):

- A **free front** moves at c ($v = 0.98$).
- With the coupling **off**, a cluster disperses (extent $28 \rightarrow 55$): unbound.
- With the coupling **on**, the cluster is **confined** (extent stays $1.4 - 2.3$) while each constituent keeps moving at c , and the composite's center of mass is **sub-luminal** ($v \approx 0.5 < c$). This is a genuine bound state of massless-moving parts.

The composite has the defining property of mass, **inertia**. Under a uniform force the bound composite responds at $v_x = 0.72$, while an *unbound* cluster of the same size responds at 0.97 ($\approx$ the free front). The resistance is from binding, not from averaging, and that gap is the controlled, load-bearing result. And the composite heads toward rest as it grows: the center-of-mass drift shrinks monotonically with

constituent count ($0.54 \rightarrow 0.31$ from eight to thirty-two), the internal momenta cancelling better in a larger bound state.

This is **mass without mass**: a bound system of c -moving constituents has rest energy, hence rest mass, and it is the physically dominant form of real mass. It is native to ED's dynamics, conditional on V5, which is a faithful structural addition (finite-reach, retarded, attractive, matching the corpus kernel) but not shown to be forced by the bare primitives.

Mass is not time dilation

The substrate has a *second* way to make something move slower than c , and it is important that it is not mass. A front carrying commitment-memory dwells, re-committing in place, which lowers its advance rate. Push such a front and its forward drift tracks its path speed exactly ($v_x/\text{path} = 1$ at every memory level): it has **no directional inertia**. A slow clock, not a mass. This is **time dilation**, and it is the same commitment-rate factor that appears in gravity's sparse-commitment parameter (§5), which is why gravitational time dilation and this share a factor. Only *binding* produces the directional inertia ($v_x/\text{path} < 1$) that is mass.

So two things the substrate had entangled come apart cleanly: **commitment sparseness sets the clock (time dilation, and it ties matter to gravity through the shared factor); binding sets the mass**. They are distinct phenomena, and the arrow is behind both, once as the commitment rate, once as what makes the bound constituents massless in the first place.

Scope

- Binding mass is native but **V5-conditional**: V5 is a structural addition, not shown primitive-forced (the bare substrate's native cross-chain coupling was separately measured to be dispersive).
- This is **binding** mass, the dominant form. The **fundamental Higgs/electroweak mass** (electron, current-quark, W/Z from spontaneous breaking) is a separate mechanism, inherited; the condensate route to it comes up empty on the certified field.
- The rest limit is an extrapolated size-trend, not a measured zero; the equivalence-principle reading of the uniform-force response is a consistent interpretation, not a proof.
- No numerical mass values, and no generations, masses, or mixings, those are the inherited value layer (§11) and the rep-spectrum content (§7, §13).

What this buys the report

The matter sector is the arrow three times over: undoubling the spinor, making the lone front massless (so mass must be binding), and setting the clock that is time dilation. The mass-from-binding result is a genuine native mechanism for the dominant form of real mass, measured on the certified substrate, and the clean mass/time-dilation split reuses the same commitment factor that runs gravity's clock, tying the matter and gravity sectors together through the arrow once more. The pieces that stay inherited, the $Cl(3,1)$ structure, the fundamental Higgs mass, the generation spectrum, all point to the same two places the report is honest about: the value layer (§11) and the rep-spectrum wall (§7, §13).

§11 — The Constants: inherited by ED’s own logic

The Standard Model has roughly twenty free parameters — the fine-structure constant, the fermion mass ratios, the mixing angles, the cosmological constant’s value, the gauge couplings — and it derives none of them. ED does not derive them either. The question this section answers is whether that is a hole in ED or a prediction of it, and the answer, argued rather than asserted, is the latter.

The structural reason

ED’s determinability boundary (A1) established a sharp fact about the substrate: the only observer-independent scalar the substrate produces in isolation is **zero**. Controlled channel capacity between regions is exactly zero (that result is the content of §8’s determinacy sibling and is derived, not posited). A substrate whose ontology is pure relation has no place to store an intrinsic dimensionless number. The corpus’s own image for this is exact: asking the substrate for the fine-structure constant is like asking one molecule for the temperature of a gas. Temperature is real, but it is a property of an ensemble, not of a fragment, and no amount of interrogating the fragment yields it.

So the constants are not quantities ED is failing to compute. They are quantities of a kind the substrate provably does not carry in isolation. The values are global, relational facts, inherited from measurement, and the framework’s own structure is what says they must be.

The taxonomy of what ED can and cannot produce

This is not a convenient after-the-fact story; it matches exactly what ED’s machinery does and does not do. Every structural move in the corpus that succeeds produces an **O(1) coefficient sitting on top of an already-inherited scale** — the holographic area count gives the $1/4$ in $S = A/4$ on top of ℓ_P ; the near-horizon smoothness gives the 2π in the Hawking temperature on top of κ ; the mixing-matrix phase count gives the combinatorial $(n-1)(n-2)/2$ with the numerical inputs still inherited. Nothing in the demonstrated toolkit produces or suppresses across orders of magnitude.

And where ED tried to derive a genuine dimensionless number, it failed, cleanly and on the record. The fine-structure constant was attempted by three independent routes (topological winding, an RG fixed point, a cross-overlap integral) and all three failed. The fermion mass ratios were attempted by six mechanisms and all six were refuted (a single real scalar field carries no topological invariant to quantize them). The cosmological constant’s magnitude has no candidate mechanism at all — standard coarse-graining produces at most polynomial corrections, not the 122 orders of magnitude required. These negatives are exactly what the structural claim predicts: ED produces forms and O(1) factors, not the values.

The honest resting place

The consistent conclusion, reached in the corpus and adopted here, is that the constants ED needs but does not fix — c , ℓ_P , $\hbar$, and the cosmological Θ_{ED} — are **substrate constants**, on the same footing as the speed of light. They are numbers the universe hands the theory, not research targets with a self-imposed moratorium. The standing directive is “think, don’t chase,” now backed by a taxonomic reason (no ED move produces this kind of number) rather than an instinct.

Two clarifications keep this honest rather than triumphant. First, this does not explain the cosmological constant; Θ_{ED} ’s magnitude is exactly as unexplained as before. Relocating it from “open target” to “substrate constant” is a bookkeeping correction, not a solution — it states plainly that ED inherits

the number, rather than pretending a primitive produces it. Second, the “no intrinsic scalar” claim is a structural posture with strong support (the A1 zero-scalar result, which is derived, plus the taxonomy of successes and failures), not a from-nothing impossibility theorem. It is as firm as A1 and the failure record make it, and it is stated at that tier.

The fair comparison

The decisive point for a skeptical reader is the same one that governs §7. The Standard Model does not derive its constants; they are its free parameters, fit to data. So ED inheriting them is not a deficiency *relative to the theory it is being compared against* — it is the identical inheritance. What ED adds on top is the structure the Standard Model leaves unexplained: the *form* of Newton’s constant ($G = c^3 \ell_P^2 / \hbar$), the *form* of the MOND scale ($a_0 = c H_0 / 2\pi$), the combinatorial *form* of the mixing-matrix phase count, the F^2 action, the single hypercharge. ED organizes the constants and derives the relations among the forms; it inherits the numbers, exactly as its ontology says it must, and exactly as the Standard Model does anyway.

What this buys the report

The constants are the report’s [inherited] column, and they belong in a different category from the two structural opens. #1 (the rep-spectrum) and #3 (the chiral anomaly cancellation) are things ED *could* still derive and has not; the constants are things ED’s own logic says it *cannot* and *need not* derive. Keeping that line sharp is what stops the twenty free parameters from reading as twenty holes. They are one principled inheritance, predicted by the substrate’s lack of an intrinsic scalar, and no worse than the Standard Model’s own standing on the same numbers.

§12 — Internal Consistency: the anomaly question, split honestly

The arrow’s job here. Two of the arrow’s results feed this section. Its parity-cleanness makes the clean substrate vector (§9), which fixes an anomaly-free baseline. And the anomaly-relevant non-Hermiticity, the thing a genuine chiral anomaly would live in, is the worldline arrow itself (a point-gap spectral flow). So the arrow both sets the safe baseline and is where the one open question lives.

Gauge anomaly cancellation is the deepest internal-consistency requirement of a chiral gauge theory: the gauge current must stay conserved at the quantum level, and in the Standard Model this works only because of a striking numerical balance among the hypercharges. The honest question is what ED contributes to this, and the answer is not “ED forces it” — an earlier version of that claim was overreached and corrected this session. The answer is that the question splits, exactly the way the chirality casting did, into a solid side and an inherited side, with one narrow open candidate.

The solid side

Two pieces are secure.

Charge conservation. The charge-as-topology result (§8) gives an exact integral Gauss law, circulation = $2\pi\omega$, loop-independent, to machine precision. An exact Gauss law is exact gauge-charge conservation, which is the substrate statement of anomaly-freedom for the charge sector. (Read with B4’s own caveat: the winding is Σ -blind and weakly coupled, so this is analytic, and reading the exact Gauss law as anomaly-freedom is an interpretive step on top of the measured conservation, taken here.)

The clean baseline. A gauge anomaly requires chiral content — left and right contribute with opposite sign, and a vector-like theory cancels automatically. §9’s theorem is that the parity-clean substrate is vector for every channel-count. So the clean substrate carries no chiral spectral flow, and there is nothing there to threaten the exact Gauss law. **The clean baseline is anomaly-free, trivially and provably** (at the transport level, contingent on the substrate-to-Dirac descent, like the theorem it rests on).

The inherited side

The Standard Model’s anomaly cancellation is *nontrivial*: the theory is chiral, and its consistency depends on the special balance $\Sigma Y = 0$ and $\Sigma Y^3 = 0$ per generation, the condition that ties the quarks (times three colors) to the leptons. This is one of the Standard Model’s most striking just-so facts, and a grand unified theory like $SO(10)$ explains it by putting a generation in a single **16**.

In ED, the chiral content appears only after the spontaneous symmetry breaking of §9, and *which* chiral content it is, is the inherited casting (the pseudoreality of $SU(2)$, the same rep-spectrum wall as §7). So the nontrivial cancellation is **inherited along with the content**. ED does not independently derive why the emergent chiral content is anomaly-free; that rides in with the representation spectrum it inherits.

The one open candidate

ED’s characteristic move is to force a *constraint*, not a *value*, and there is exactly one place it might do so here. B4’s Gauss law is *exact*. If that exact charge conservation must survive the arrow’s chiral spectral flow, then the substrate would **forbid** a gauge anomaly by construction — forcing *that* whatever chiral content emerges is anomaly-free, without ever selecting *which* content. That would be a genuine, non-inherited ED contribution to the deepest consistency condition in particle physics.

It is a candidate, not a result, and it is gated. Establishing it requires the substrate-to-Dirac worldline reduction (T4, §10), because a gauge anomaly is properly a relativistic object and the spectral-flow face lives there. The operator triage is complete — the anomaly-relevant non-Hermiticity is specifically the worldline arrow, and every off-worldline candidate (the V1 kernel, V5, the Lindblad effective Hamiltonian) was checked and ruled out for a distinct reason — so the question is correctly located, but the reduction that would answer it is the one deep arc still open in the matter sector. The candidate is named, not claimed.

The correction, on the record

This section exists partly to state a retraction plainly. An earlier note this session argued that ED *forces* anomaly-freedom, by reading B4’s classical winding-protection as quantum anomaly-freedom. That conflated two different things: an anomalous theory also has a classically conserved current, so classical winding-protection does not discriminate anomaly-free from anomalous content. Adversarial review caught it, and it contradicted Paper_CleanSubstrateVector §6 from the same day. The claim was downgraded to the candidate above. The reason to foreground this rather than bury it is that it is the report’s honesty discipline working as designed: the retraction is why the anomaly box is a gated candidate and not a false [check].

Scope

- The conservation face and the clean baseline are solid; the nontrivial chiral cancellation is inherited; the “constraint-forcing” possibility is a T4-gated candidate.
- All of it is at the transport/structural level; the relativistic anomaly proper is gated on the substrate-to-Dirac reduction.
- ED does not derive the Standard Model’s anomaly cancellation. The section maps the question to its honest floor.

What this buys the report

The anomaly box is the report’s second genuine structural open (#3), and it is smaller and better-located than #1. Its solid pieces are real (exact conservation, a provably vector baseline), its inherited piece is the same rep-spectrum wall the rest of the Standard Model quarter inherits, and its one non-inherited possibility is precisely gated on the T4 reduction that also gates §9’s relativistic chirality and §10’s graph-native spinor. So the two deep matter-sector arcs, the anomaly constraint-face and the substrate-to-Dirac descent, are one arc, named honestly and left open.

§13 — The Walls: what ED does not reach, and of how many kinds

A unified framework is judged as much by the sharpness of its stated limits as by its results. ED’s limits are not one undifferentiated “not yet.” They come in distinct kinds, and reporting them at the same flat tier would be its own dishonesty. There are four: one proven wall, two bounded structural opens, a set of principled inheritances, and (separately, §11) the constants. This section states the first three plainly.

The one proven wall: primality

ED has exactly one limit that is a *theorem*, and it is worth stating first because it is the strongest evidence that the report’s walls are real and not rhetorical.

ED’s substrate is, by its primitives, in the finite-reach / finite-memory class: every determination is a function of a bounded local residue state, which is a periodic, zero-entropy sequence. Using the integers as an external ruler, the prime structure splits into a **template** (the finite-local residue statistics — coverage by small primes, equidistribution among open residues) and an **escape** (primality itself, the Möbius sign, pair correlations). The finite-memory class reproduces the template exactly — the scale-invariant 1.700-bit template invariant and the least-prime-factor density ladder are matched to five decimals — and **provably cannot reach the escape**. The optimal finite-memory correlator with the Möbius function decays to zero for every fixed memory ($\sim\sqrt{M/N}$), the per-open-position primality entropy is irreducible, and the twin density requires a pair-correlation the template does not supply. The mechanism is a $\sqrt{N}$ activation horizon: the memory needed to resolve more grows with the scale it must resolve, so no fixed finite memory ever catches up.

What makes this a *proven* wall rather than a located one is that it is anchored to established theorems outside ED — Möbius orthogonality to periodic sequences (the prime-number theorem in arithmetic progressions) and the sieve parity barrier — not to ED grading its own homework. ED belongs to the class those theorems constrain, so the ceiling is located by theorem.

Two things keep this honest. It is not a failure: ED is not in the business of generating the primes, and the integers are a ruler, not a target. And it is a strength of the report, not a weakness — a framework

that can name exactly one thing it provably cannot do, and cite the theorem, is demonstrating that its other walls are drawn with the same seriousness. This wall is also the external twin of A1 (§8's determinacy sibling): the same finite-reach ceiling, measured once internally (controlled channel capacity is exactly zero) and once against a math-certified ruler.

The two structural opens (located, not proven)

Two limits are genuine open problems — things ED *could* still derive or delegate, not things proven impossible. These are the report's real [open]'s.

#1, the representation spectrum. Why the stable channel-families are $\{1,2,3\}$, and behind it, why the internal amplitude dimension is three. ED's natural route (channel-family stability) was tested and refuted, and that is reported as a failure (§7). What remains is bounded: the question reduces to a single sharp one (why internal-d = 3), and its one live route turns on a standard, non-ED-specific computational task — detecting whether the participation graph is intrinsically linked, which needs an established linkless-embedding-detection algorithm, not new ED insight. It is delegatable. And it is unsolved in the Standard Model too, which postulates the multiplicities.

#3, the chiral anomaly cancellation. Whether ED's exact Gauss law forces anomaly-freedom on the emergent chiral content, ED's one possible non-inherited contribution to the deepest consistency condition in the Standard Model (§12). It is gated on the substrate-to-Dirac worldline reduction (T4), which is also what gates §9's relativistic chirality and §10's graph-native spinor. So the two structural opens are really *one deep arc* — the substrate-to-Dirac reduction — plus the rep-spectrum tooling task. Two well-located problems, not a diffuse fog.

The principled inheritances (settled, not open)

Two further limits are neither proven walls nor open problems: they are resolved as inherited, with reasons.

Dimension. Three-dimensionality (P06) is selected, not derived. But it is selected independently three ways — the internal amplitude dimension from the gauge content, the holographic reach law from observed gravity, and a linking argument from the substrate's own topology — and the linking route is a genuine derivation *candidate* that would route 3D back through the arrow (committed order held stable by linking, which is unique to three dimensions). So dimension is inherited with a conditional route home, not a blank.

Casting. Which force is chiral is resolved-as-inherited: the pseudoreality of $SU(2)$'s fundamental versus the complex representations of $SU(N \geq 3)$, a representation-theoretic fact ED carries (§9). It is not an open ED derivation and not a wall to break; it is the same rep-spectrum inheritance as #1, correctly located outside ED's dynamical output.

The map of “no”

So ED's limits sort cleanly, and the sorting is the honest picture:

- **Proven wall (1):** primality — theorem-anchored, the finite-memory ceiling. ED knows exactly one thing it provably cannot do.
- **Structural opens (2):** the rep-spectrum (#1, delegatable tooling task) and the anomaly constraint (#3, gated on substrate-to-Dirac). Both bounded, both unsolved in the Standard Model, both on the same two arcs.

- **Principled inheritances:** dimension (selected, conditional route home) and casting (re-theoretic, resolved-as-inherited).
- **Inherited-by-design (§11):** the constants — a fourth category, things ED’s own logic says it cannot and need not derive.

None of these is hidden, and none is dressed as more settled than it is. That is the point of the section: the walls are drawn honestly and they are of different kinds, and the report’s credibility comes from keeping those kinds distinct rather than collapsing them into a single apologetic “open.”

What this buys the report

The walls set up the two sections that close the report. §14 asks the harder question — not what ED has not yet derived, but what could prove it *wrong* — and §15 states the bottom line. The honest map here is what lets those sections be read as confident rather than defensive: a framework that has located its one proven limit, bounded its two real opens, and named its inheritances has earned the right to state plainly what it does and does not claim.

§14 — The Falsifier Frontier: what could prove ED wrong

A framework earns the word “unified” by explaining, and it earns the word “scientific” by being killable. This section is the second test. It asks where ED diverges sharply enough from Λ CDM, standard quantum mechanics, and general relativity to be proven wrong at a measurable point, and it states the honest status of the one thing that would end the argument: **no distinctive, argument-ending prediction is confirmed yet.**

What counts as a weapon

The corpus draws a line that this report keeps. A **weapon** is a prediction that is *novel* (standard physics does not make the claim), *sharp* (a number, a sign, or a shape, not a direction), *near-term testable*, and *distinctive* (ED and the standard account diverge at a measurable point). A **postdiction** — reproducing an already-known value — is consilience: it supports the framework but cannot end an argument, and several such postdictions are shared with Λ CDM, MOND, or GR. An internal-consistency check is not an empirical discriminator at all. Because ED inherits the value layer, no weapon here is a predicted *number* from first principles; the weapon is always the *form, sign, or shape*, and that is what can fail.

The ranked weapons

These are ED’s sharpest falsifiable bets, in the corpus’s own priority order.

1. **The participation-envelope ratio.** ED predicts a universal ratio $f_v \approx 1.1 \cdot \gamma_{\text{dec}}$ (the Q/π form, corrected in 2026-07-10 from an earlier erroneous $8 \cdot \gamma_{\text{dec}}$) between a slow coherence-envelope modulation and the decoherence rate, holding across more than ten orders of magnitude in decoherence rate. No standard theory predicts a universal ratio spanning optomechanics to biological FRAP; decoherence is supposed to be monotonic with no slow envelope. The detector is built and validated; the accessible band is $\sim 11\text{--}110$ Hz (a FRAP Lomb–Scargle measurement). Commissioned FRAP is currently unfunded, so the live zero-cost paths are a literature-data dig or the spatial-PDE harmonic legs.

2. **The merging-cluster knee.** ED predicts a sharp, step-like knee in the offset-velocity relation of merging clusters (flat $\rightarrow$ knee $\rightarrow$ linear $\rightarrow$ ceiling), with fast mergers showing more than two lensing peaks. Λ CDM predicts velocity-*independent* scatter (no knee); MOND predicts a *smooth* roll-off (no knee). The ceiling and the >2 -peak leg are testable now on El Gordo-class mergers; the full knee awaits Rubin/Euclid. The shape is robust; the knee's location is inherited.
3. **The Class-A decoherence wall.** ED predicts a sharp, architecture-*independent* mass wall for quantum superposition at 140–250 kDa, with a few-percent second-harmonic fringe near the crossing. Standard decoherence is smooth and environment-dependent, with no universal wall. Matter-wave interferometry is at ~ 25 kDa and climbing.
4. **The error-correction plateau.** ED predicts logical error floors that do not go to zero ($\Gamma_{\text{plateau}} > 0$) and a specific exponential rigidity in the approach. Standard fault tolerance predicts error $\rightarrow 0$ exponentially with code distance. Industry is climbing code distance now, so this is a distinctive test arriving essentially for free.
5. **BTFR zero intrinsic scatter.** ED predicts *exactly zero* intrinsic scatter in the deep-MOND asymptote of the baryonic Tully–Fisher relation. Λ CDM halo models predict nonzero scatter from formation-history variance. This is, in the corpus's words, the cleanest single number to miss, and it is consistent with SPARC data so far.
6. **Dark energy pinned to $w = -1$.** ED predicts saturation, not quintessence: the dark-energy equation of state is pinned to $w = -1$ with no evolution. Quintessence and phantom models allow $w \neq -1$ and $w_a \neq 0$. DESI and Euclid are measuring this now; if the hint of evolution firms up, it is a near-term kill — which is exactly what makes it a real weapon.
7. **$\alpha_2 = 0$ exactly, and the matter-sector prohibitions.** The preferred-frame parameter α_2 is identically zero (§5), where a generic khronometric or Lorentz-violating theory gives $\alpha_2 \sim O(1)$; this is a kill-switch ED passes (GR passes it too, so it is a survival, not a win over GR). And the matter sector carries hard prohibitions from the clean-vector theorem (§9): no parity-violating chiral $U(1)$ force, no stable gauge group beyond $SU(3)$, no anyons in genuine 3+1D. Any counterexample is a one-line kill, and competitors do not make these prohibitions.

Several of these trace directly to arrow-sector results: $\alpha_2 = 0$ and the pinned dark energy come from the khronon (§5), and the matter-sector prohibitions come from the clean-vector theorem (§9). The arrow that unifies the report is also where several of its sharpest falsifiers come from.

The confirmed floor

Two predictions are settled, and it is important to state exactly what they are worth.

- **The Universal Degenerate-Mobility law** $D(c) = D_0(1 - c/c_{\text{max}})^\beta$ is peer-reviewed (May 2026), confirmed across ten materials at $R^2 > 0.986$. This is the corpus's one genuine *confirmed forward* prediction — a real result, and the floor the report stands on.
- **$c_{\text{GW}} = c$** (GW170817, to $< 10^{-15}$) is a survival that killed many rival modified-gravity theories. But ED inherits it structurally (§5), so it is consilience, not a distinctive win.

Neither of these is the argument-ending weapon. The mobility law is a soft-matter scaling law that other frameworks can accommodate; the GW speed is a survival ED shares with GR. They establish that ED makes correct forward predictions; they do not, by themselves, single ED out.

The honest north-star

The plain status: **none of the distinctive, argument-ending weapons is confirmed.** The envelope ratio, the cluster knee, the Class-A wall, and the BTFR zero-scatter are the bets that would single ED out, and all are unconfirmed. And the recent foundational work is honest about its own limits here: the quantum-logic reconstruction (§4) and the chiral-gauge sector (§9) closed structure and located walls, but they produced *no new near-term weapon*. So the highest-leverage move toward the goal is not further derivation; it is executing the Tier-1 bets above.

Two distinctive claims are explicitly *not yet* weapons, and the report flags them as such rather than inflating them. Primitive einselection and the emergent multi-context structure (§4) are genuinely distinctive but have no concrete near-term observable. And the gauge-handedness/matter-sign correlation (§9) is theoretically firm but is flagged in the register as not testable with current data, so it belongs here as a distinctive-but-not-yet-testable claim, not a live weapon.

What this buys the report

This is where the report is most honest and most exposed, and deliberately so. ED is falsifiable at many sharp, distinctive points — a universal envelope ratio, a step-like cluster knee, an architecture-independent decoherence wall, an exactly-zero BTFR scatter, a pinned dark-energy equation of state, a set of one-line matter-sector prohibitions. Its credibility as a unified framework ultimately does not rest on the derivations in §3–§13. It rests on confirming one of these, and that has not happened yet. The report’s final section states what that means.

§15 — The Honest Bottom Line

Event Density is **form-complete and not closed**, and both halves of that sentence are meant precisely.

What it is

ED classifies the *form* of every sector a unified framework must address, and it does so from one primitive. The arrow of commitment reconstructs quantum mechanics by selecting the pointer basis and supplying measurement repeatability (§4); it closes gravity as the khronon, a genuine preferred time, unifying the weak-field Einstein metric with dark matter and dark energy in one field (§5); and, most distinctively, those two jobs are the *same* arrow, so the problem of time dissolves rather than being reconciled (§6). Around that spine, the Standard Model sectors are the same primitive in smaller roles: it quantizes and protects charge and casts electromagnetism as a coarse-grained shadow (§8), it is half of the chirality operator and forces parity violation to be spontaneous (§9), it undoubles the spinor and makes a lone excitation massless so that mass comes from binding (§10), and it anchors the single hypercharge of the derived gauge structure (§7). The values that fill in these forms — the couplings, the masses, the mixing angles, the cosmological constant — are inherited, and inherited *by principle*: a pure-relation substrate produces no intrinsic scalar, so it cannot and need not set a dimensionless number (§11), exactly as the Standard Model inherits the same twenty parameters.

That is a serious and coherent claim. One primitive, worked independently across sectors that mainstream physics treats as unrelated, keeps turning up as the mechanism, most strikingly where quantum mechanics and gravity were supposed to be irreconcilable.

What is open

The report does not close, and it is exact about where it stops.

Two structural questions are genuinely unbuilt. Why the gauge multiplicities are $\{1,2,3\}$ (#1, §7) and whether ED's exact charge conservation forces anomaly-freedom on the emergent chiral content (#3, §12). These are not a diffuse fog; they reduce to **one deep arc and one tooling task** — the substrate-to-Dirac worldline reduction (which also gates the relativistic chirality statement and the graph-native spinor), and a delegatable linkless-embedding computation that would test whether the internal dimension is three. Both are unsolved in the Standard Model too, which postulates what ED is trying to derive.

One wall is proven: the finite-memory substrate provably cannot reach the primality escape layer (§13), theorem-anchored, the program's one certified negative. And the remaining limits — three-dimensionality, the chirality casting, the constants — are principled inheritances, not open derivations.

What is not done

Most importantly: **no distinctive, argument-ending prediction is confirmed** (§14). ED makes one confirmed forward prediction (the degenerate-mobility law) and survives the tests that killed rival theories ($c_{\text{GW}} = c$), but the bets that would single it out — the universal envelope ratio, the step-like cluster knee, the architecture-independent decoherence wall, the exactly-zero BTFR scatter — are all still open. Until one of them confirms, ED is a coherent, falsifiable unified framework, not a certified one. That distinction is the honest ceiling of this entire document, and it is stated as plainly as it can be.

What would close it

Three things, in rough order of decisiveness:

1. **A confirmed novel prediction.** The decisive one. Executing the near-term Tier-1 bets (§14) is the highest-leverage work toward it, and the corpus is explicit that further derivation is *not* — the recent foundational work closed structure but produced no new weapon.
2. **The substrate-to-Dirac reduction (T4).** This single arc gates the relativistic chirality statement, the graph-native spinor, and the one non-inherited anomaly candidate. Closing it would convert three of the matter sector's "form-complete, structure-inherited" results into deeper derivations.
3. **The linkless-embedding test.** A bounded, delegatable computation that would either derive the internal dimension (un-walling #1) or confirm it is honestly inherited.

The bottom line

ED is a unified framework in the strong sense — one primitive, the arrow of time, doing the load-bearing work across quantum mechanics, gravity, and the Standard Model's structure — that is form-complete, value-inherited by its own logic, honest about two genuine structural opens and one proven wall, and not yet confirmed at any of its distinctive falsifiers. The case it makes is that the recurrence of a single primitive across sectors is real and worth taking seriously; the case it does not yet make is the empirical one that would end the argument. This document carries both openly, because a unified-framework report that hid its opens and its unconfirmed status would be dismissible, and one that states them at their honest tier is not.

Appendices

Appendix A — Paper index by box

Each box of the report is delivered by one or more corpus papers. The cold-reader papers (in physics-papers/, ED-Generative repo) are the primary citations — the ones a physicist would open. Where a result is newer than its published paper, the working note carrying it (in event-density/foundations/ or /theory/) is named and flagged.

§	Box	Authoritative paper(s)	Location
§3	The arrow / minimal foundation	<i>Commitment and Participation (Minimal Ontology); The Arrow Sorts the Continuum; the position paper (Paper_087_13Primitive)</i>	physics-papers/substrate-evaluation/; physics-papers/foundations/
§4	Quantum mechanics	<i>The Quantum-Logic Keystone (Gleason Reconstruction); Paper_004 (the postulated inner product it reconstructs); Paper_001 (C-amplitude)</i>	physics-papers/substrate-evaluation/; physics-papers/qm-kinematics/
§5	Gravity (GR + DM + DE)	<i>GR-I..IV; KM-I/II; One Field (letter); Papers 025–038 (G, a_0, combination rule, BTFR, MOND field eqn, Λ). λ-bound: working note Khronometric_Lambda_HealthyBranch_FromStandingKeystone</i>	physics-papers/gravity/; event-density/foundations/ (λ note)
§6	QM + gravity unified	<i>Commitment and Participation §8 (the keystone synthesis + its “account, not theorem” guard)</i>	physics-papers/substrate-evaluation/
§7	Gauge structure	<i>The Gauge Structure of ED ($SU(N)$, F^2, mass gap, single hypercharge)</i>	physics-papers/substrate-evaluation/
§8	Charge + electromagnetism	<i>The Topological Skeleton of Charge (B4); Maxwell as an Emergent Shadow; The Continuum: a Kinetic Lattice-Gas</i>	physics-papers/substrate-evaluation/

§	Box	Authoritative paper(s)	Location
§9	Chirality / parity	<i>The Clean Substrate Is Vector</i> (supersedes <i>The Parity Wall</i> on the verdict's tier)	physics-papers/substrate-evaluation/
§10	Matter (spinor + mass)	<i>Mass Without Mass (Binding Inertia)</i> ; Paper_106 (the Dirac equation); working note T4_14_Closure_SubstrateToDirac (form-completeness + undoubling)	physics-papers/substrate-evaluation/, physics-papers/relativistic-event-density/foundations/ (T4 note)
§11	The constants	<i>Common Cause, Not Channel (A1)</i> (the zero-scalar result); working notes SCBU_SubstrateEvaluationDensityFoundations/, Scop-ing_ThetaED_FirstPrinciples working note	physics-papers/substrate-evaluation/; event-density/foundations/, /docs/
§12	Anomalies	Anomaly_State_After_Clean_Verify_Theor_Amo_2026-07-10; B4 (conservation face); <i>Clean Substrate Is Vector</i> §6 (baseline)	event-density/foundations/, physics-papers/substrate-evaluation/
§13	The walls	<i>Template, Not Escape (Primes)</i> (the one proven wall); A1; gauge paper §7 (#1); ontology paper §6 (dimension); clean-vector §5 (casting)	physics-papers/substrate-evaluation/
§14	Falsifier frontier	<i>Paper_101 — Falsification Register and Prediction Inventory</i> §0 (weapons-first, 2026-07-10); ED_Master_Predictions_List	physics-papers/predictions/; event-density/docs/

Navigation aids in the EDG repo: *PAPERS_INDEX.md* (the full corpus index) and each folder's *README.md* (per-arc tiering). The authoritative open/closed map is *event-density/docs/ED_Research_Targets.md*.

Appendix B — Provenance and honesty ledger

The report's honesty tiers are not set by each paper's own framing; they are set against the research map. This is the condensed ledger — the full audit trail (with dated updates) is event-density/docs/ED_Open_Derivations_Ledger.md and ED_Research_Targets.md.

The five derivations tracked this session, at close:

#	Derivation	Status at close
1	Channel-topology → representation spectrum	Open (tooling-walled). Reduces to “why internal-d = 3”; ED's stability route refuted; the one lead (3D-linking bridge) needs a published linkless-embedding algorithm. Delegatable; unsolved in the SM too.
2	Mass / electroweak / Higgs	Honest-closed. Native binding-mass mechanism (measured, V5-conditional); k_{11} separated as time-dilation, not mass; fundamental Higgs inherited.
3	Anomaly-freedom as a forced constraint	Structural candidate, gated. Conservation + clean-vector baseline solid; the “forced” claim retracted; nontrivial chiral cancellation inherited; one candidate gated on substrate→Dirac.
4	DCGT → Maxwell	Done. Coherence-weighted limit = Coulomb (computed); smooth field an emergent shadow.
5	Exact khronometric λ	Done (bound) / inherited (value). Healthy branch $\lambda < 1/3$, $\lambda \neq 1$ derived; exact λ inherited.

The map of “no” (from §13): - **Proven wall (1):** primality — theorem-anchored (Möbius orthogonality + parity barrier), the finite-memory ceiling. - **Structural opens (2):** #1 (rep-spectrum, delegatable tooling task) and #3 (anomaly, gated on substrate→Dirac) — one deep arc plus one tooling task. - **Principled inheritances:** dimension (selected, conditional linking route home); casting (SU(2) pseudoreality). - **Inherited-by-design:** the constants (no intrinsic scalar; A1).

Confirmation status: one confirmed forward prediction (the Universal Degenerate-Mobility law, peer-reviewed); $c_{GW} = c$ survived (consilience); **no distinctive, argument-ending weapon confirmed** — the honest ceiling.

Appendix C — The certified simulations

The “measured” tier means a result was produced by a built-and-run simulation of the certified substrate (or, for the analytic probes, a direct computation of the stated quantity). These are the runnable artifacts behind the report’s measured claims; paths are in the ED-Generative and event-density repos.

Result (§)	Script	What it runs
Charge skeleton, integral Gauss law (§8)	holonomy_test.py, coupling_test.py, sourcing_test.py, relaxation_test.py	B4: winding quantization to $\mathbb{Z}$; w-indexed bandwidth ladder; circulation = $2\pi w$ loop-independent; the ontology fork (1/r ² only off-ED) — evaluation/B4_Arc/
Maxwell shadow (§8)	maxwell_from_coherence_probe.py	For Poisson solve of the coherence-action minimizer around a point charge; p=1 (3D Coulomb) fit R ² =0.97 — evaluation/B4_Arc/
QM orthogonality + covering law (§4)	move1_operational_orthogonality_glean_nonboolean_probe.py	For distinguishingability $\Leftrightarrow$ orthogonality (1-c ²); the non-Boolean complementarity gate — evaluation/ChiralGauge/
Chirality: clean-vector theorem (§9)	rep_spectrum_casting_winding.py	For clean point-gap winding = 0 for N=1..6 (control on the theorem); broken-case winding = N — evaluation/ChiralGauge/
Spinor undoubling (§10)	chiral_3p1d.py	Nielsen–Ninomiya: Hermitian naive = 16 doublers, arrow (Wilson term) = 1 survivor at the origin — evaluation/ChiralGauge/
Binding mass + inertia (§10)	mass_from_binding_probe.py	free front v=0.98; unbound vs V5-confined (extent 55 vs 1.4–2.3); COM sub-luminal; inertia 0.72 vs unbound-control 0.97; the k ₁₁ time-dilation discriminator — theory/Higgs_Emergence/
Khronon dynamical rule (§5)	GR-III simulation ($\ddot{b} = D\nabla^2 b - \kappa\rho$)	Newtonian fixed point; r _s $\propto$ M; frozen b→0 horizon + area law; khronon speed c _s = c — gravity arc
Channel capacity = 0 (§8, §11, §13)	A1 channel-capacity probe	controlled capacity exactly zero (the internal reading of the finite-reach ceiling)

Result (§)	Script	What it runs
Finite-memory ceiling (§13)	<code>primes_test.py</code>	sieve to $N=5 \times 10^6$: template reproduced (1.700-bit invariant, lpf ladder); escape blocked (optimal Möbius correlator $\rightarrow 0$; twin density needs $2C_2$) — evaluation/Primes_Arc/

Verification status (2026-07-10): every script path above was confirmed present. Four were re-run this session and reproduce their cited numbers exactly — `rep_spectrum_casting_winding.py` (clean winding $0 \forall N$, broken = N), `chiral_3p1d.py` (16 doublers $\rightarrow$ 1 survivor at the origin), `maxwell_from_coherence_probe.py` ($p=1$ 3D-Coulomb, $R^2=0.97$), and `mass_from_binding_probe.py` (inertia 0.72 vs unbound-control 0.97). The remainder (the B4 holonomy suite, `move1/gLeason`, GR-III, A1, `primes_test.py`) are path-verified with numbers as last run in their source papers; re-run before publication for a clean provenance stamp.

Honest scope note: the gauge sector's non-abelian action and mass-gap (§7) are analytic (gauge-program tier), not certified-simulator runs — only the abelian/Maxwell case is simulator-grounded. The chirality theorem's winding computation is a control on a proof, not an independent measurement. Both are flagged as such in their sections.